

Toward Downscaled Estuarine Salinity: Evidence from Alpine Catchments

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Abstract

Work in environmental research has long relied on assumptions we test directly here. The reduced model retains most of the explanatory power at a fraction of the computational cost. Residual error grows approximately linearly with the number of coupled parameters. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

1. Introduction

The reduced model retains most of the explanatory power at a fraction of the computational cost. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Longer observation windows narrow the confidence interval but do not shift the central estimate. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. Neither of the alternative formulations improved on the simple additive model in our setting.

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The measurements reported here were collected over twelve independent trials under controlled conditions. Taken together, these findings argue for treating the two regimes separately in future analyses.

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. Longer observation windows narrow the confidence interval but do not shift the central estimate. We release the processing scripts and intermediate artefacts to support independent verification.

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2. Related Work

Taken together, these findings argue for treating the two regimes separately in future analyses. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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3. Materials and Methods

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4. Experimental Setup

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5. Results

We release the processing scripts and intermediate artefacts to support independent verification. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting. The reduced model retains most of the explanatory power at a fraction of the computational cost. The measurements reported here were collected over twelve independent trials under controlled conditions. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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6. Discussion

Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate. Calibration drift over the trial period remained below the threshold set out in the protocol. The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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7. Limitations

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